AMINO ACIDS

OBJECTIVES

- Describe diagrammatically the general structure of an Amino Acid with example.
- Give a general classification of Amino Acids with examples and biomedical significance of each type.
- What is meant by Hydrophobic and hydrophilic Amino Acids ? Give examples
- Describe all Essential and Non essential Amino Acids with their types, biomedical and clinical significance.
- What is meant by Glucogenic and ketogenic Amino Acid? Differentiate them and give examples of both types with their biomedical significance.
- How does degradation , removal and excretion of Amino Acids take place
- Discuss the degradative fate of the following Amino Acids
- Glutamine (b) Histidine (c) Proline (d) Arginine

- Enumerate those Amino Acids which yield Oxaloacetate and α-Ketoglutarate, dicuss their respective pathways with their biomedical and clinical significance.

- Discuss the formation of Pyruvate and Fumarate from Amino Acids with examples. And significance.

- Discuss the pathways for the synthesis of methionine , S- Adenosyl methionine and cysteine with their biomedical importance

- Enlist the Amino Acids that yield Succinyl CoA and Acetyl CoA

- What is oxidative decarboxylation?. Describe in detail with examples

What is meant by one carbon metabolism? What is its relation to AA ?Give its significance

- How are Non- essential Amino Acids formed ?

- Discuss the following clinical disorders related to AA metabolism

Phenylketonuria b) Homocystinuria c) Alkaptonuria d)Maple syrup urine diseas

Introduction

- **Definition:**

"The building blocks of proteins are known as Amino Acids"

OR

"A simple organic compound containing both a carboxyl (—COOH) and an amino ($—NH_2$) group is called Amino Acid"

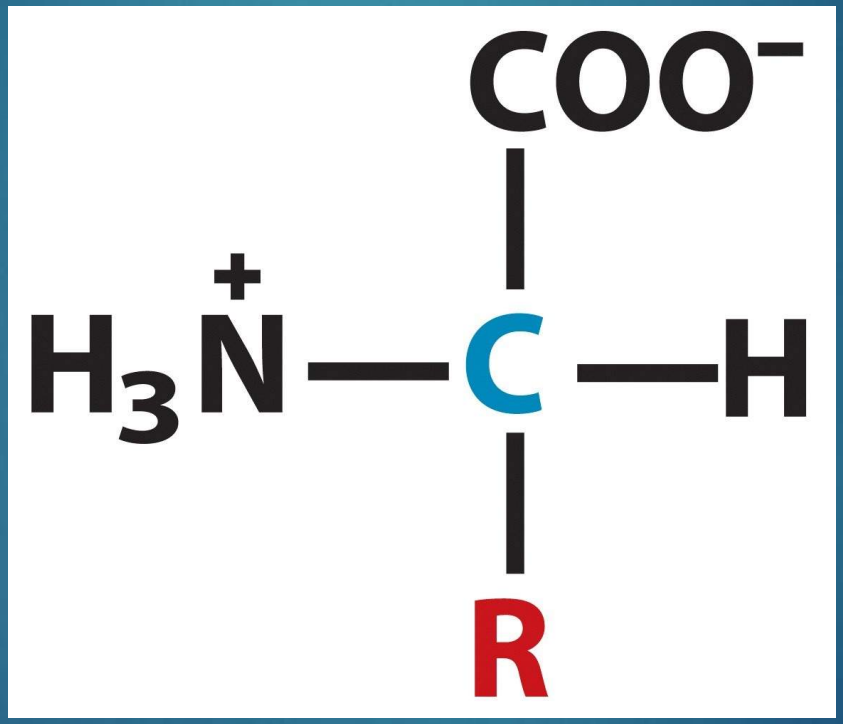
Introduction

- Essential components of peptides and proteins.
- Used for a protein synthesis
- There are Standard and Non Standard Amino Acids
- They are described on the basis of side chains

Standard Amino Acids

- They take part in protein synthesis
- More then 300 AA have been described
- Only 20 AA are found in mammalian tissue
- These 20 AA are called primary or standard AA
- There are 9 essential and 11 non essential AA

Standard Amino Acids



Standard Amino Acids

- Each AA contain a amino group (NH2), carboxyl group (COOH) and a distinct side chain
- Exception to this rule is **Proline** which contains ***Imino*** group (NH) instead of ***Amino*** group

Standard Amino Acids

Phenylalanine	Histidine
Tryptophan	Arginine
Valine	Lysine
Threonine	Leucine
Leucine	Alanine
Isoleucine	Cysteine
Methionine	Glutamic Acid
Serine	Aspartic Acid

Non Standard Amino Acids

- These Amino acids do not take part in the protein synthesis but play important role in the body.

Citrulline
Ornithine
Taurine
DOPA
GABA

Classification of Amino Acids

- **Amino Acids are classified on the basis of side chains into:**

- Amino acids with Aromatic side chain
- Amino acids with aliphatic side chain
- Amino acids with side chain containing Sulphur atom
- Amino acids with Acidic side chains
- Amino acids with Basic side chains
- Amino acids with side chain having OH group
- Imino Acid

Classification of Amino Acids

Amino acids with Acidic Side Chains

- These are mono amino dicarboxylic
- Aspartic acid
- Glutamic acid

Amino acids with Basic Side Chains

- These are diamino monocarboxylic
- Arginine
- Histidine
- Lysine

Classification of Amino Acids

Amino Acids with Aliphatic Side Chain

- Glycine
- Alanine
- Valine
- Leucine
- Isoleucine

Amino Acids with Side Chain Having OH Group

- Serine
- Threonine

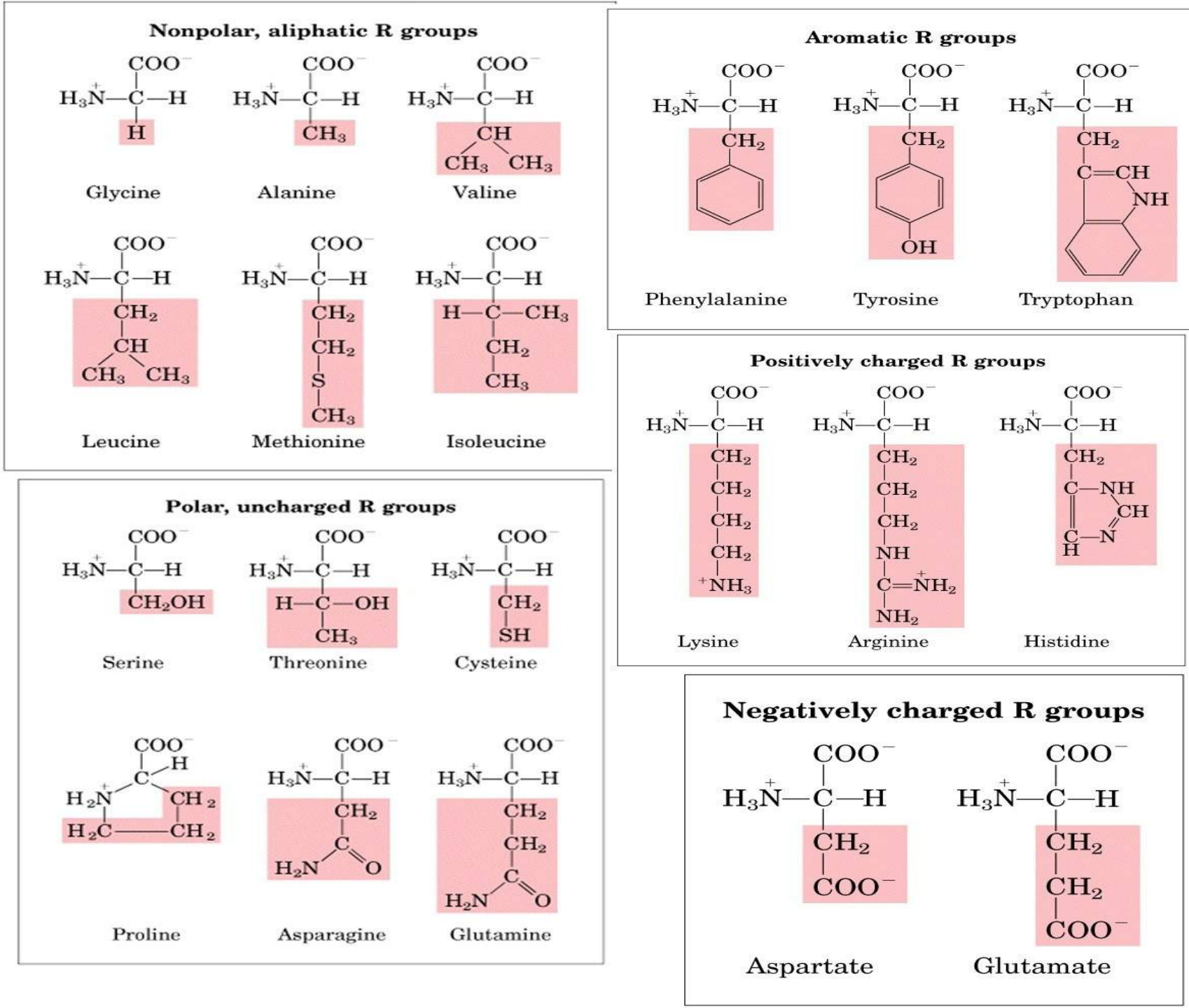
Classification of Amino Acids

Amino Acids with Side Chain Containing Sulphur Atom

- Cysteine
- Cystine
- Methionine

Amino Acids with Aromatic Side Chain

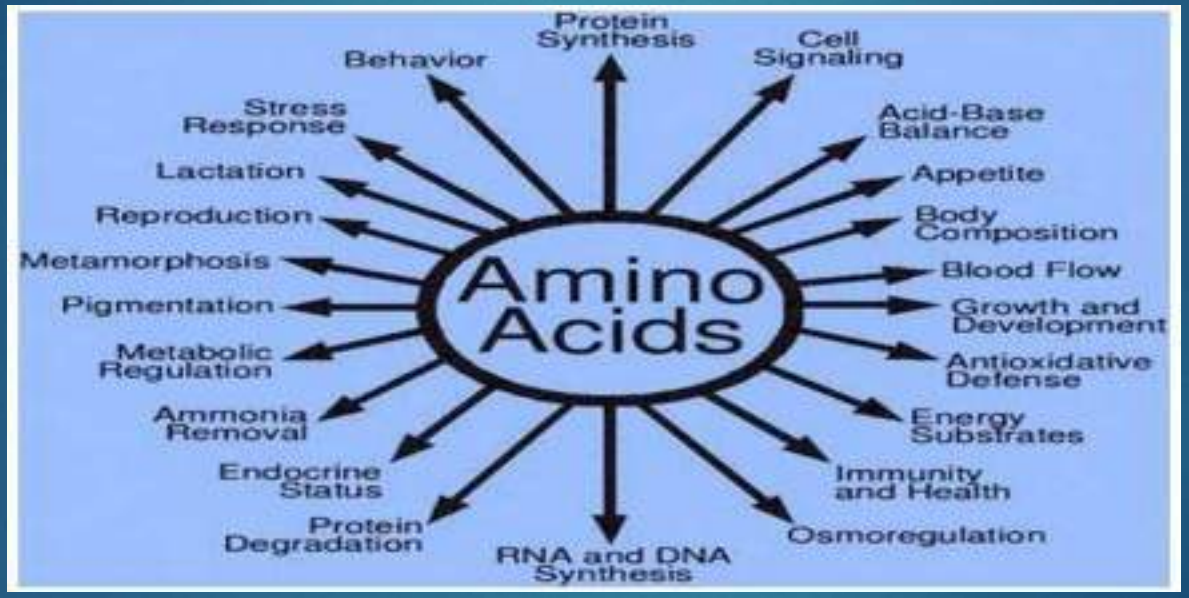
- Phenylalanine
- Tyrosine
- Tryptophan


Nonpolar, aliphatic R groups
Glycine
Alanine
Valine
Leucine
Methionine
Isoleucine
Aromatic R groups
Phenylalanine
Tyrosine
Tryptophan
Positively charged R groups
Lysine
Arginine
Histidine
Polar, uncharged R groups
Serine
Threonine
Cysteine
Proline
Asparagine
Glutamine
Negatively charged R groups
Aspartate
Glutamate

Functions of Amino Acids

- They play variety of roles in metabolism:
- They are building blocks of proteins
- They form part of coenzymes
- They act as precursor for the biosynthesis of molecules. Eg: Haem
- They are a source of energy; like proteins, they can provide about 4 Calories per gram, etc

Functions of Amino Acids



Biomedical Importance of AA

- **Phenylalanine**: It is a precursor for neurotransmitters tyrosine, dopamine, epinephrine and norepinephrine. It plays an integral role in the structure and function of proteins and enzymes and the production of other amino acids

- **Valine:** It helps to stimulate muscle growth and regeneration and is involved in energy production

Biomedical Importance of AA

- **Histidine:** It is used to produce histamine, a neurotransmitter that is vital to immune response, digestion, sexual function and sleep-wake cycles. It's critical for maintaining the myelin sheath, a protective barrier that surrounds your nerve cells

- **Tryptophan:** It's needed to maintain proper nitrogen balance and is a precursor to serotonin, a neurotransmitter that regulates appetite, sleep and mood

Biomedical Importance of AA

- **Methionine:** It plays an important role in metabolism and detoxification. It's also necessary for tissue growth and the absorption of zinc and selenium, minerals that are vital to health

Food Sources of Amino Acids

- Essential Amino Acids cannot be produced by body so they must be provided through diet
- Foods that contain all nine essential amino acids are referred to as complete proteins.

Complete protein sources include:

- Meat
- Seafood
- Poultry

Food Sources of Amino Acids

- Eggs
- Dairy Products
- Soy
- Quinoa
- Buckwheat

Health Benefits of Amino Acids

- Prevent muscle loss
- Increase muscle growth
- Decrease muscle soreness
- Reduce exercise fatigue
- Benefit people with Liver diseases
- Promote weight loss
- Improve mood

Nitrogen Balance

- Nitrogen balance is the (normal) condition in which the amount of nitrogen
- incorporated into the body each day exactly equals the amount excreted.
- Negative nitrogen balance occurs when nitrogen loss exceeds incorporation. It
- is associated with:
- • Protein malnutrition (kwashiorkor)
- • Dietary deficiency of even 1 essential amino acid
- • Starvation
- • Uncontrolled diabetes
- • Infection

- Positive nitrogen balance occurs when the amount of nitrogen incorporated
- exceeds the amount excreted. It is associated with:
- • Growth
- • Pregnancy
- • Convalescence (recovery phase of injury or surgery)
- • Recovery from condition associated with negative nitrogen balance

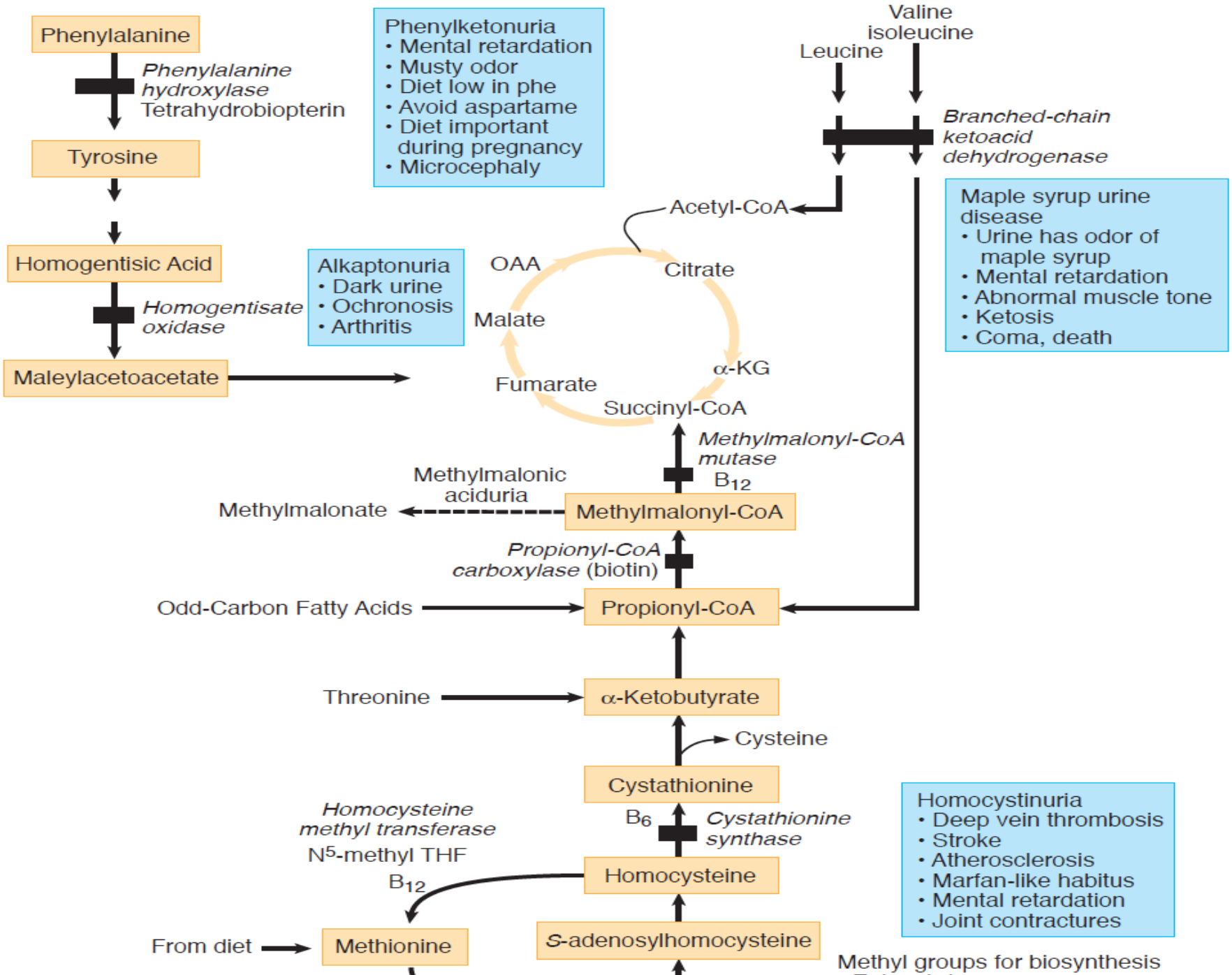
AMINO ACID METABOLISM

Protein obtained from the diet or from body protein during prolonged fasting or starvation may be used as an energy source. Body protein is catabolized primarily in muscle and in liver. Amino acids released from proteins usually lose their amino group through transamination or deamination. The carbon skeletons can be converted in the liver to glucose (glucogenic amino acids), acetyl CoA, and ketone bodies (ketogenic), or in a few cases both may be produced (glucogenic and ketogenic)

REMOVAL AND EXCRETION OF AMINO GROUPS

- Excess nitrogen is eliminated from the body in the urine. The kidney adds small quantities of ammonium ion to the urine in part to regulate acid-base balance, but nitrogen is also eliminated in this process. Most excess nitrogen is converted to urea in the liver and goes through the blood to the kidney, where it is eliminated in urine.
- Amino groups released by deamination reactions form ammonium ion (NH_4^+),which must not escape into the peripheral blood. An elevated concentration of ammonium ion in the blood, hyperammonemia, has toxic effects in the brain(cerebral edema, convulsions, coma, and death). Most tissues add excess nitrogen to the blood as glutamine by attaching ammonia to the γ-carboxyl group of glutamate. Muscle sends nitrogen to the liver as alanine and smaller quantities of other amino acids, in addition to glutamine.

DISORDERS OF AMINOACIDS METABOLISM


Phenylalanine
Phenylalanine hydroxylase
Tetrahydrobiopterin
Tyrosine
Homogentisic Acid
Homogentisate oxidase
Maleylacetoacetate
Phenylketonuria
• Mental retardation
• Musty odor
• Diet low in phe
• Avoid aspartame
• Diet important during pregnancy
• Microcephaly
Alkaptonuria
• Dark urine
• Ochronosis
• Arthritis
Valine
isoleucine
Leucine
Branched-chain ketoacid dehydrogenase
Maple syrup urine disease
• Urine has odor of maple syrup
• Mental retardation
• Abnormal muscle tone
• Ketosis
• Coma, death
Acetyl-CoA
OAA
Citrate
Malate
α-KG
Fumarate
Succinyl-CoA
Methylmalonyl-CoA mutase
B12
Methylmalonic aciduria
Methylmalonate
Methylmalonyl-CoA
Propionyl-CoA carboxylase (biotin)
Odd-Carbon Fatty Acids
Propionyl-CoA
Threonine
α-Ketobutyrate
Cysteine
Cystathionine
Homocysteine methyl transferase
N5-methyl THF
B12
B6
Cystathionine synthase
Homocysteine
Homocystinuria
• Deep vein thrombosis
• Stroke
• Atherosclerosis
• Marfan-like habitus
• Mental retardation
• Joint contractures
From diet
Methionine
S-adenosylhomocysteine
Methyl groups for biosynthesis


Thank
you!